D Y PATIL UNIVERSITY

RAMRAO ADIK INSTITUTE OF TECHNOLOGY

NAVI MUMBAI

Questions Engineering Physics

Module 1: Interference and Diffraction:

1. Why do we observe different colors in thin films in sun light? With neat diagram, derive an expression for optical path difference) in interference in thin films of constant thickness due to reflected light. And hence deduce the conditions for constructive and destructive interference
2. What do you mean by thin film? Derive an expression for Optical path difference due to interference of monochromatic ray of reflected light by thin film of constant thickness. Also deduce condition for constructive and destructive interference.
3. What is the nature of interference fringes in wedge shaped film and why? Derive an expression for bandwidth.
4. Show that interference fringes in wedge shaped film due to reflected light are equidistant. Consider case of normal incidence.
5. What is Anti-Reflection Coatings? Discuss it in details with phase and amplitude conditions.
6. What do you mean by antireflection coatings? Discuss antireflection coatings in details in terms of path difference, thickness, and refractive indices for thin film.
7. Why fringes in Newton's Rings experiment are Circular? Show that diameter of dark rings are proportional to square root of order/integer?
8. Explain determination of wavelengths of visible spectra using plane diffraction grating.
9. Why Newton's rings are circular in nature? A Newton's ring setup is illuminated by a beam of light containing two wavelengths; 4500 Å and 4000 Å. It is found that nth dark ring due to one wavelength coincides with (n+3)th dark ring of another wavelength. If radius of curvature of the lens is 57 cm then find order and radius of nth dark ring?
10. In Newton's ring experiment the diameter of 5th and 10th dark ring are 3.5 mm and 6.4 mm respectively. Find the diameter of 15th dark ring. If a liquid is introduced between Plano-convex lens and glass plate then new diameter of 15th dark ring appears to be 5.45 mm. Calculate refractive index of liquid.

11. In a wedge shaped air film, 30 dark fringes are observed within 3.2 mm spacing. If wavelength of incident light is 5890 Å , find the wedge angle in degree, minutes and seconds
12. How many orders will be visible if the wavelength of incident radiation is 5000 Å and the number of lines on the grating is 2620 in one inch..?
13. White light is incident on soap film at an angle sin-1(4/5) and reflected light after examination by spectroscope showed dark bands. Two consecutive dark bands are observed at wavelengths 6100 Å and 6000 Å, if the refractive index of the film is 4/3, what will be the thickness of the film?
14. White light falls normally on a soap film of refractive index 1.5 and thickness 4500 Å. Which wavelengths within the visible spectrum (λ = 4000 A0 to 7000 A0) will be strongly reflected?
15. A plane transmission grating having 5000 lines per cm is used to obtain a source of light with the third order. Calculate the angular separation between the spectral lines whose wavelengths are 500 and 515 nm.
16. A diffraction grating used at normal incidence gives a line 4500 Å in certain order superimposed on another line 4120 Å of next higher order. If the angle of diffraction is 45º, find the no. of lines per cm on the grating.

Module 2 LASERS and FIBER OPTICS

17. Explain stimulated emission and hence explain working of laser with help of its components.
18. Differentiate between spontaneous emission and stimulated emission.
19. Write short note on Holography.
20. Define acceptance angle, fractional index change of an optical Fiber. Derive an expression for Numerical Aperture of Step Index Fiber. Discuss the significance of numerical aperture.
21. Explain phenomenon of Spontaneous, stimulated emission and Population inversion. With the help of neat labelled diagrams explain the construction and working of He-Ne Laser.
22. Explain phenomenon of stimulated emission, metastable state, Optical Pumping. With the help of neat labelled diagrams explain the construction and working of Nd:YAG Laser.
23. Differentiate Step Index Fiber and Graded Index Fiber. Draw Profile index Diagram for Step Index Fiber and Graded Index Fiber.

24. With help of block diagram, explain optical fiber communication system. What are the advantages and disadvantages of optical fiber communication system over wired and wireless communication?
25. What do you mean by modes in optical fiber? Explain different type of modes travelling through fiber in short.
26. A light ray enters an optical fiber of core refractive index 1.49 and cladding refractive index 1.44 Find the following:
27. Critical Angle at core-cladding interface
28. Numerical Aperture
29. V Number and Number of modes travelling through fiber of core radius 90 μm and 750 nm is wavelength of light.

Module 3 QUANTUM MECHANICS

1. Show that de Broglie group velocity associated with the wave packet is equal to the particle velocity.
2. What do you mean by matter wave? Explain Properties of matter waves.
3. Using Heisenberg's Uncertainty Principle, show that electron cannot exist within the nucleus.
4. Derive one dimensional Time Dependent Schrodinger equation for matter waves.
5. Derive one dimensional Schrodinger's Time Independent wave equation.
6. Show that the energy of an electron in the box varies as the square of natural number.
7. Derive the expression for energy Eigen values for a free particle in one dimensional potential well.
8. Show that energy levels of a particle in one dimensional infinite potential well are quantized.
9. State and Explain de Broglie Hypothesis. An electron beam is accelerated from rest through a potential difference of 200 V. Calculate the associated wavelength.
10. Calculate de Broglie wavelength associated with an $\alpha-$particle accelerated by a potential difference of 100 kV. Mass of $\alpha-$particle is 6.68×10^{-27}kg.

11. An electron is trapped in a one dimensional box of length 0.1 nm. Calculate the energy required to excite the electron from its ground state to the second excited state.
12. A position and momentum of 1 KeV electron are simultaneously measured. If position is located within 10 nm then what is the percentage uncertainty in its momentum?
13. The speed of an electron is measured to within an uncertainty of 2 x 10^4 m/s. What is the minimum space required by the electron to be confined to an atom?

Module 4 SEMICONDUCTORS

1. Derive microscopic form of Ohm's law. Deduce conductivity for an intrinsic semiconductor.
2. What do you mean by Fermi energy? Explain it with Fermi Dirac distribution function.
3. Show that Fermi energy lie at the midway of band gap in intrinsic semiconductor.
4. What is effect of Temperature and doping concentration on Fermi Energy in N type semiconductor? Explain with necessary diagram.
5. What is effect of Temperature and doping concentration on Fermi Energy in P type semiconductor? Explain with necessary diagram.
6. Write a short note on LED with suitable diagrams.
7. Write short note on photovoltaic cell with suitable diagrams.
8. The resistivity of Cu is 1.72 × 10-8 Ohm -m. Calculate the mobility of electrons in Cu. Given number of electrons per unit volume is 10.41 x 1028 / m3
9. Find resistivity of intrinsic germanium at 330 K Given density of carriers is 2.5 x 1019 / m3. Mobility of electrons is 0.39 m2/V sec. and for holes 0.19 m2/V sec.
10. Find out probability of electron not occupying in Energy level E in conduction band at 273 K for pure Si of Band gap 0.74 eV.
11. Find out probability of electron occupying energy level E lying above Fermi energy by 0.25 eV energy at 300K.
12. Find out probability of electron occupying energy level E lying below Fermi energy by 0.13 eV energy at 300K.

Module 5 Nanotechnology

1. What do you mean by nanomaterials? How nanoparticles are differ from their bulk form in terms of properties?
2. Explain following properties of nanoparticles in details: SVR, Blue-shift, Superparamagnetism, Quantum confinement.
3. Explain Top Down and Bottom up approaches for the synthesis of Nanomaterials with examples.
4. What do you mean by bottom up and top down approach to synthesize nanomaterials? Write short note on ball milling method.
5. Explain Principle and working of Atomic Force Microscope (AFM) with suitable diagram.
6. Explain Principle and working of Scanning Electron Microscope (SEM) with suitable diagram.
7. What are the applications of nanomaterials?

Module 6: PHYSICS OF SENSORS:

1. How transducers differ from sensors? Explain concept of sensitivity, selectivity, accuracy and precision.
2. Define Hall Effect. Derive an expression of Hall Voltage. Explain physical significance of Hall coefficient.
3. Why photodiode is used in reversed bias for light sensing? Explain construction and working of Photodiode with I-V characteristics.
4. A sample of n type silicon has donar density of 10^{21}/m3. It is used in Hall effect experiment. If the sample of width 0.64 mm is kept in a magnetic field of 5500 Gauss with current density 400 A/m2. Calculate Hall potential, Hall Coefficient and Hall Angle (Given:Mobility is 0.17 m2/Vsec.)